

# ECEN 629 Convex Optimization for Electrical Engineering

## Chapter 3. Convex Functions

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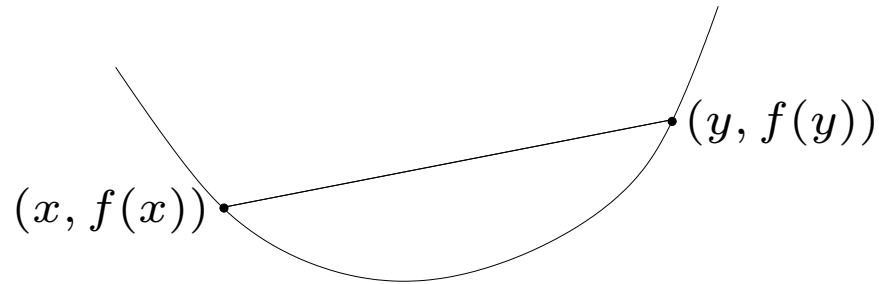
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## Definition

$f : \mathbf{R}^n \rightarrow \mathbf{R}$  is convex if  $\mathbf{dom} f$  is a convex set and

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

for all  $x, y \in \mathbf{dom} f$ ,  $0 \leq \theta \leq 1$



- $f$  is concave if  $-f$  is convex
- $f$  is strictly convex if  $\mathbf{dom} f$  is convex and

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

for  $x, y \in \mathbf{dom} f$ ,  $x \neq y$ ,  $0 < \theta < 1$

## Examples on $\mathbf{R}$

convex:

- affine:  $ax + b$  on  $\mathbf{R}$ , for any  $a, b \in \mathbf{R}$
- exponential:  $e^{ax}$ , for any  $a \in \mathbf{R}$
- powers:  $x^\alpha$  on  $\mathbf{R}_{++}$ , for  $\alpha \geq 1$  or  $\alpha \leq 0$
- powers of absolute value:  $|x|^p$  on  $\mathbf{R}$ , for  $p \geq 1$
- negative entropy:  $x \log x$  on  $\mathbf{R}_{++}$

concave:

- affine:  $ax + b$  on  $\mathbf{R}$ , for any  $a, b \in \mathbf{R}$
- powers:  $x^\alpha$  on  $\mathbf{R}_{++}$ , for  $0 \leq \alpha \leq 1$
- logarithm:  $\log x$  on  $\mathbf{R}_{++}$

## Examples on $\mathbf{R}^n$ and $\mathbf{R}^{m \times n}$

affine functions are convex and concave; all norms are convex

### examples on $\mathbf{R}^n$

- affine function  $f(x) = a^T x + b$
- norms:  $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$  for  $p \geq 1$ ;  $\|x\|_\infty = \max_k |x_k|$

### examples on $\mathbf{R}^{m \times n}$ ( $m \times n$ matrices)

- affine function

$$f(X) = \mathbf{tr}(A^T X) + b = \sum_{i=1}^m \sum_{j=1}^n A_{ij} X_{ij} + b$$

- spectral (maximum singular value) norm

$$f(X) = \|X\|_2 = \sigma_{\max}(X) = (\lambda_{\max}(X^T X))^{1/2}$$

## Restriction of a convex function to a line

$f : \mathbf{R}^n \rightarrow \mathbf{R}$  is convex if and only if the function  $g : \mathbf{R} \rightarrow \mathbf{R}$ ,

$$g(t) = f(x + tv), \quad \text{dom } g = \{t \mid x + tv \in \text{dom } f\}$$

is convex (in  $t$ ) for any  $x \in \text{dom } f$ ,  $v \in \mathbf{R}^n$

can check convexity of  $f$  by checking convexity of functions of one variable

**example.**  $f : \mathbf{S}^n \rightarrow \mathbf{R}$  with  $f(X) = \log \det X$ ,  $\text{dom } f = \mathbf{S}_{++}^n$

$$\begin{aligned} g(t) = \log \det(X + tV) &= \log \det X + \log \det(I + tX^{-1/2}VX^{-1/2}) \\ &= \log \det X + \sum_{i=1}^n \log(1 + t\lambda_i) \end{aligned}$$

where  $\lambda_i$  are the eigenvalues of  $X^{-1/2}VX^{-1/2}$

$g$  is concave in  $t$  (for any choice of  $X \succ 0$ ,  $V$ ); hence  $f$  is concave

## Extended-value extension

extended-value extension  $\tilde{f}$  of  $f$  is

$$\tilde{f}(x) = f(x), \quad x \in \mathbf{dom} f, \quad \tilde{f}(x) = \infty, \quad x \notin \mathbf{dom} f$$

often simplifies notation; for example, the condition

$$0 \leq \theta \leq 1 \quad \implies \quad \tilde{f}(\theta x + (1 - \theta)y) \leq \theta \tilde{f}(x) + (1 - \theta) \tilde{f}(y)$$

(as an inequality in  $\mathbf{R} \cup \{\infty\}$ ), means the same as the two conditions

- $\mathbf{dom} f$  is convex
- for  $x, y \in \mathbf{dom} f$ ,

$$0 \leq \theta \leq 1 \quad \implies \quad f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

# First-order condition

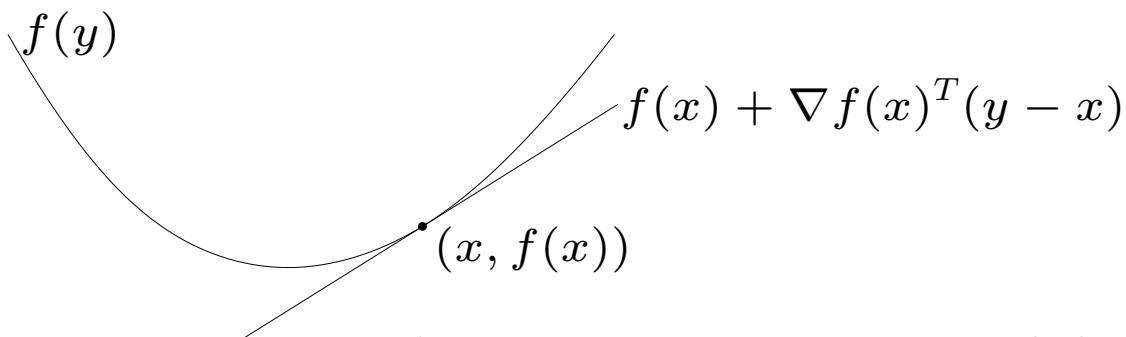
$f$  is **differentiable** if  $\text{dom } f$  is open and the gradient

$$\nabla f(x) = \left( \frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right)$$

exists at each  $x \in \text{dom } f$

**1st-order condition:** differentiable  $f$  with convex domain is convex iff

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \text{dom } f$$



first-order approximation of  $f$  is global underestimator

## Second-order conditions

$f$  is **twice differentiable** if  $\text{dom } f$  is open and the Hessian  $\nabla^2 f(x) \in \mathbf{S}^n$ ,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \dots, n,$$

exists at each  $x \in \text{dom } f$

**2nd-order conditions:** for twice differentiable  $f$  with convex domain

- $f$  is convex if and only if

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \in \text{dom } f$$

- if  $\nabla^2 f(x) \succ 0$  for all  $x \in \text{dom } f$ , then  $f$  is strictly convex



# Global Properties and Strict Convexity

From the first order condition  $f(y) \geq f(x) + \nabla f(x)^T(y - x)$

- From local properties to global properties;
- When  $\nabla f(x)^T = 0$  then for all  $y \in \mathbf{dom}f$ ,  $f(y) \geq f(x)$ .

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Strict convexity

- If and only if  $\mathbf{dom} f$  is convex and for any  $x, y \in \mathbf{dom} f$ ,  $x \neq y$

$$f(y) > f(x) + \nabla f(x)^T(y - x).$$

# Global Properties and Strict Convexity

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Domain being convex is important

- Consider  $f(x) = 1/x^2$  with  $\mathbf{dom} f = \{x \in \mathbf{R} | x \neq 0\}$ .
- $f''(x) > 0$ , but it is not convex.

## Examples

**quadratic function:**  $f(x) = (1/2)x^T P x + q^T x + r$  (with  $P \in \mathbf{S}^n$ )

$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

convex if  $P \succeq 0$

**least-squares objective:**  $f(x) = \|Ax - b\|_2^2$

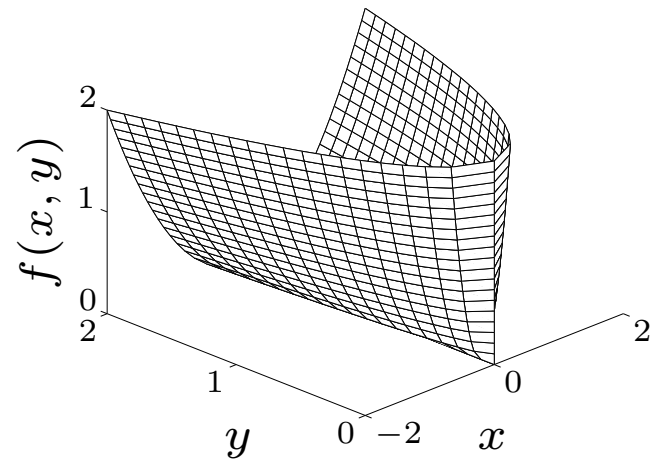
$$\nabla f(x) = 2A^T(Ax - b), \quad \nabla^2 f(x) = 2A^T A$$

convex (for any  $A$ )

**quadratic-over-linear:**  $f(x, y) = x^2/y$

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{bmatrix} y \\ -x \end{bmatrix} \begin{bmatrix} y \\ -x \end{bmatrix}^T \succeq 0$$

convex for  $y > 0$



**log-sum-exp:**  $f(x) = \log \sum_{k=1}^n \exp x_k$  is convex

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z} \mathbf{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T \quad (z_k = \exp x_k)$$

to show  $\nabla^2 f(x) \succeq 0$ , we must verify that  $v^T \nabla^2 f(x) v \geq 0$  for all  $v$ :

$$v^T \nabla^2 f(x) v = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k v_k z_k)^2}{(\sum_k z_k)^2} \geq 0$$

since  $(\sum_k v_k z_k)^2 \leq (\sum_k z_k v_k^2)(\sum_k z_k)$  (from Cauchy-Schwarz inequality)

**geometric mean:**  $f(x) = (\prod_{k=1}^n x_k)^{1/n}$  on  $\mathbf{R}_{++}^n$  is concave

(similar proof as for log-sum-exp)

# Epigraph and sublevel set

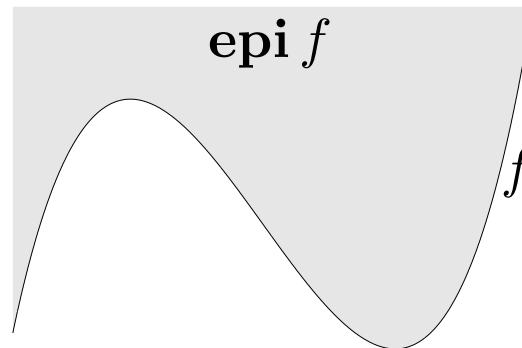
$\alpha$ -sublevel set of  $f : \mathbf{R}^n \rightarrow \mathbf{R}$ :

$$C_\alpha = \{x \in \mathbf{dom} f \mid f(x) \leq \alpha\}$$

sublevel sets of convex functions are convex (converse is false)

**epigraph** of  $f : \mathbf{R}^n \rightarrow \mathbf{R}$ :

$$\mathbf{epi} f = \{(x, t) \in \mathbf{R}^{n+1} \mid x \in \mathbf{dom} f, f(x) \leq t\}$$



$f$  is convex if and only if  $\mathbf{epi} f$  is a convex set

# Sublevel Sets and Convex Functions

**Sublevel sets of a convex function are convex:**

- If  $x, y \in C_\alpha$ , then  $f(x) \leq \alpha$  and  $f(y) \leq \alpha$

$$f(\theta x + (1 - \theta)y) \leq \alpha \Rightarrow \theta x + (1 - \theta)y \in C_\alpha.$$

# Sublevel Sets and Convex Functions

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- Can be used to establish a set is convex. E.g. fix  $\alpha \in [0, 1]$

$$\left\{ x \in \mathbf{R}_+^n \mid \left( \prod_{i=1}^n x_i \right)^{1/n} \geq \alpha \frac{1}{n} \sum_{i=1}^n x_i \right\}.$$



# Sublevel Sets and Convex Functions

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**A non-convex function can have all its sublevel sets convex**

- Example:  $f(x) = -e^x$ .

# Jensen's inequality

**basic inequality:** if  $f$  is convex, then for  $0 \leq \theta \leq 1$ ,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

**extension:** if  $f$  is convex, then

$$f(\mathbf{E} z) \leq \mathbf{E} f(z)$$

for any random variable  $z$

basic inequality is special case with discrete distribution

$$\mathbf{prob}(z = x) = \theta, \quad \mathbf{prob}(z = y) = 1 - \theta$$

# Inequalities

**Many known inequalities are derived through convexity:**

- Arithmetic-geometric mean inequality  $\sqrt{ab} \leq (a + b)/2$  for  $a, b \geq 0$ .

$$f(x) = -\log x \text{ is convex} \Rightarrow -\log \left( \frac{a + b}{2} \right) \leq \left( \frac{-\log a - \log b}{2} \right)$$

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- Hölder's inequality: for  $p > 1$ ,  $1/p + 1/q = 1$  and  $x, y \in \mathbf{R}^n$

$$\sum x_i y_i \leq \left( \sum |x_i|^p \right)^{1/p} \left( \sum |y_i|^q \right)^{1/q}$$

# Operations that preserve convexity

practical methods for establishing convexity of a function

1. verify definition (often simplified by restricting to a line)
2. for twice differentiable functions, show  $\nabla^2 f(x) \succeq 0$
3. show that  $f$  is obtained from simple convex functions by operations that preserve convexity
  - nonnegative weighted sum
  - composition with affine function
  - pointwise maximum and supremum
  - composition
  - minimization
  - perspective

# Positive weighted sum & composition with affine function

**nonnegative multiple:**  $\alpha f$  is convex if  $f$  is convex,  $\alpha \geq 0$

**sum:**  $f_1 + f_2$  convex if  $f_1, f_2$  convex (extends to infinite sums, integrals)

**composition with affine function:**  $f(Ax + b)$  is convex if  $f$  is convex

## examples

- log barrier for linear inequalities

$$f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x), \quad \text{dom } f = \{x \mid a_i^T x < b_i, i = 1, \dots, m\}$$

- (any) norm of affine function:  $f(x) = \|Ax + b\|$

## Pointwise maximum

if  $f_1, \dots, f_m$  are convex, then  $f(x) = \max\{f_1(x), \dots, f_m(x)\}$  is convex

### examples

- piecewise-linear function:  $f(x) = \max_{i=1,\dots,m}(a_i^T x + b_i)$  is convex
- sum of  $r$  largest components of  $x \in \mathbf{R}^n$ :

$$f(x) = x_{[1]} + x_{[2]} + \dots + x_{[r]}$$

is convex ( $x_{[i]}$  is  $i$ th largest component of  $x$ )

proof:

$$f(x) = \max\{x_{i_1} + x_{i_2} + \dots + x_{i_r} \mid 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$$

## Proof: Pointwise Maximum is Convex

$$\begin{aligned} f(\theta x + (1 - \theta)y) &= \max\{f_1(\theta x + (1 - \theta)y), f_2(\theta x + (1 - \theta)y)\} \\ &\leq \max\{\theta f_1(x) + (1 - \theta)f_1(y), \theta f_2(x) + (1 - \theta)f_2(y)\} \\ &\leq \theta \max\{f_1(x), f_2(x)\} + (1 - \theta) \max\{f_1(y), f_2(y)\} \\ &= \theta f(x) + (1 - \theta)f(y). \end{aligned}$$



# Pointwise supremum

if  $f(x, y)$  is convex in  $x$  for each  $y \in \mathcal{A}$ , then

$$g(x) = \sup_{y \in \mathcal{A}} f(x, y)$$

is convex

## examples

- support function of a set  $C$ :  $S_C(x) = \sup_{y \in C} y^T x$  is convex
- distance to farthest point in a set  $C$ :

$$f(x) = \sup_{y \in C} \|x - y\|$$

- maximum eigenvalue of symmetric matrix: for  $X \in \mathbf{S}^n$ ,

$$\lambda_{\max}(X) = \sup_{\|y\|_2=1} y^T X y$$

# Composition with scalar functions

composition of  $g : \mathbf{R}^n \rightarrow \mathbf{R}$  and  $h : \mathbf{R} \rightarrow \mathbf{R}$ :

$$f(x) = h(g(x))$$

$f$  is convex if  $\begin{array}{l} g \text{ convex, } h \text{ convex, } \tilde{h} \text{ nondecreasing} \\ g \text{ concave, } h \text{ convex, } \tilde{h} \text{ nonincreasing} \end{array}$

- proof (for  $n = 1$ , differentiable  $g, h$ )

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$$

- note: monotonicity must hold for extended-value extension  $\tilde{h}$

## examples

- $\exp g(x)$  is convex if  $g$  is convex
- $1/g(x)$  is convex if  $g$  is concave and positive

# Vector composition

composition of  $g : \mathbf{R}^n \rightarrow \mathbf{R}^k$  and  $h : \mathbf{R}^k \rightarrow \mathbf{R}$ :

$$f(x) = h(g(x)) = h(g_1(x), g_2(x), \dots, g_k(x))$$

$f$  is convex if  $\begin{array}{l} g_i \text{ convex, } h \text{ convex, } \tilde{h} \text{ nondecreasing in each argument} \\ g_i \text{ concave, } h \text{ convex, } \tilde{h} \text{ nonincreasing in each argument} \end{array}$

proof (for  $n = 1$ , differentiable  $g, h$ )

$$f''(x) = g'(x)^T \nabla^2 h(g(x)) g'(x) + \nabla h(g(x))^T g''(x)$$

## examples

- $\sum_{i=1}^m \log g_i(x)$  is concave if  $g_i$  are concave and positive
- $\log \sum_{i=1}^m \exp g_i(x)$  is convex if  $g_i$  are convex

# Composition

Extended-value extension  $\tilde{h}$  condition is important:

- Example:  $g(x) = x^2$  and  $\text{dom } g = \mathbf{R}$  and  $h(x) = 0$  with  $\text{dom } h = [1, 2]$

$$h(g(x)) = 0, \text{dom } f = [-\sqrt{2}, -1] \cup [1, \sqrt{2}].$$

# Minimization

if  $f(x, y)$  is convex in  $(x, y)$  and  $C$  is a convex set, then

$$g(x) = \inf_{y \in C} f(x, y)$$

is convex

## examples

- $f(x, y) = x^T A x + 2x^T B y + y^T C y$  with

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0, \quad C \succ 0$$

minimizing over  $y$  gives  $g(x) = \inf_y f(x, y) = x^T (A - B C^{-1} B^T) x$

$g$  is convex, hence Schur complement  $A - B C^{-1} B^T \succeq 0$

- distance to a set:  $\mathbf{dist}(x, S) = \inf_{y \in S} \|x - y\|$  is convex if  $S$  is convex

## Minimization over a Convex Set

**Proof:** take an arbitrary  $\epsilon > 0$

$$\begin{aligned} g(\theta x_1 + (1 - \theta)x_2) &= \inf_{y \in C} f(\theta x_1 + (1 - \theta)x_2, y) \\ &\leq f(\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2) \\ &\leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2) \\ &\leq \theta g(x_1) + (1 - \theta)g(x_2) + \epsilon. \end{aligned}$$

# Perspective

the **perspective** of a function  $f : \mathbf{R}^n \rightarrow \mathbf{R}$  is the function  $g : \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$ ,

$$g(x, t) = tf(x/t), \quad \text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}$$

$g$  is convex if  $f$  is convex

## examples

- $f(x) = x^T x$  is convex; hence  $g(x, t) = x^T x/t$  is convex for  $t > 0$
- negative logarithm  $f(x) = -\log x$  is convex; hence relative entropy  $g(x, t) = t \log t - t \log x$  is convex on  $\mathbf{R}_{++}^2$
- if  $f$  is convex, then

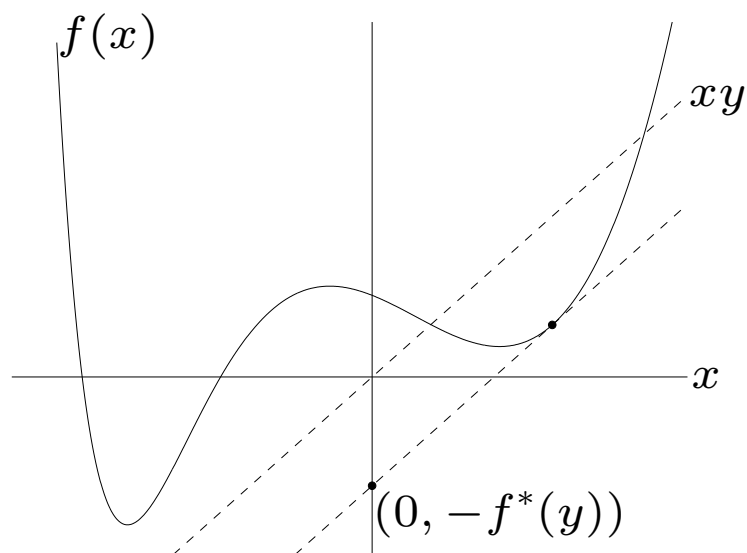
$$g(x) = (c^T x + d)f((Ax + b)/(c^T x + d))$$

is convex on  $\{x \mid c^T x + d > 0, (Ax + b)/(c^T x + d) \in \text{dom } f\}$

# The conjugate function

the **conjugate** of a function  $f$  is

$$f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$$



- $f^*$  is convex (even if  $f$  is not)
- will be useful in chapter 5



## examples

- negative logarithm  $f(x) = -\log x$

$$\begin{aligned} f^*(y) &= \sup_{x>0} (xy + \log x) \\ &= \begin{cases} -1 - \log(-y) & y < 0 \\ \infty & \text{otherwise} \end{cases} \end{aligned}$$

- strictly convex quadratic  $f(x) = (1/2)x^T Qx$  with  $Q \in \mathbf{S}_{++}^n$

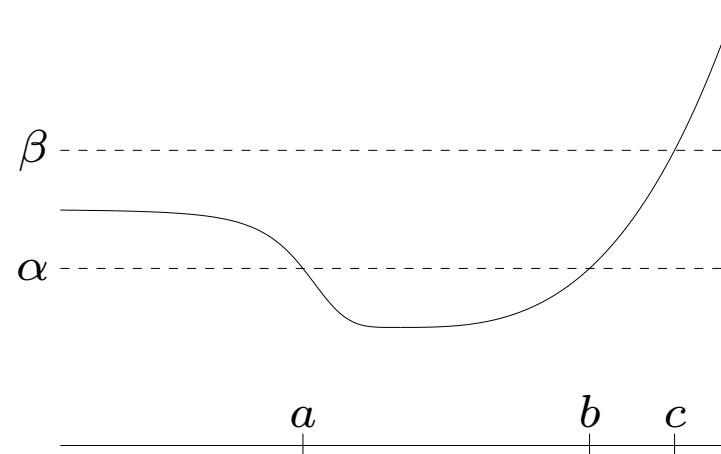
$$\begin{aligned} f^*(y) &= \sup_x (y^T x - (1/2)x^T Qx) \\ &= \frac{1}{2} y^T Q^{-1} y \end{aligned}$$

# Quasiconvex functions

$f : \mathbf{R}^n \rightarrow \mathbf{R}$  is quasiconvex if  $\mathbf{dom} f$  is convex and the sublevel sets

$$S_\alpha = \{x \in \mathbf{dom} f \mid f(x) \leq \alpha\}$$

are convex for all  $\alpha$



- $f$  is quasiconcave if  $-f$  is quasiconvex
- $f$  is quasilinear if it is quasiconvex and quasiconcave

# Examples

- $\sqrt{|x|}$  is quasiconvex on  $\mathbf{R}$
- $\text{ceil}(x) = \inf\{z \in \mathbf{Z} \mid z \geq x\}$  is quasilinear
- $\log x$  is quasilinear on  $\mathbf{R}_{++}$
- $f(x_1, x_2) = x_1 x_2$  is quasiconcave on  $\mathbf{R}_{++}^2$
- linear-fractional function

$$f(x) = \frac{a^T x + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}$$

is quasilinear

- distance ratio

$$f(x) = \frac{\|x - a\|_2}{\|x - b\|_2}, \quad \text{dom } f = \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$$

is quasiconvex

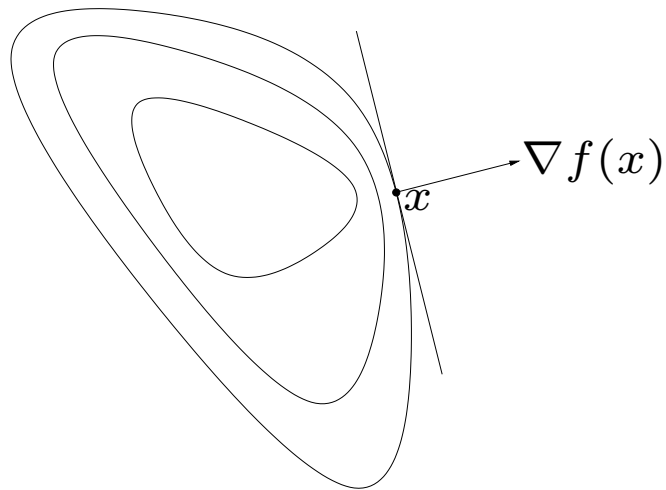
# Properties

**modified Jensen inequality:** for quasiconvex  $f$

$$0 \leq \theta \leq 1 \implies f(\theta x + (1 - \theta)y) \leq \max\{f(x), f(y)\}$$

**first-order condition:** differentiable  $f$  with cvx domain is quasiconvex iff

$$f(y) \leq f(x) \implies \nabla f(x)^T (y - x) \leq 0$$



**sums** of quasiconvex functions are not necessarily quasiconvex

## Log-concave and log-convex functions

a positive function  $f$  is log-concave if  $\log f$  is concave:

$$f(\theta x + (1 - \theta)y) \geq f(x)^\theta f(y)^{1-\theta} \quad \text{for } 0 \leq \theta \leq 1$$

$f$  is log-convex if  $\log f$  is convex

- powers:  $x^a$  on  $\mathbf{R}_{++}$  is log-convex for  $a \leq 0$ , log-concave for  $a \geq 0$
- many common probability densities are log-concave, *e.g.*, normal:

$$f(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-\bar{x})^T \Sigma^{-1}(x-\bar{x})}$$

- cumulative Gaussian distribution function  $\Phi$  is log-concave

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du$$

# Properties of log-concave functions

- twice differentiable  $f$  with convex domain is log-concave if and only if

$$f(x)\nabla^2 f(x) \preceq \nabla f(x)\nabla f(x)^T$$

for all  $x \in \text{dom } f$

- product of log-concave functions is log-concave
- sum of log-concave functions is not always log-concave
- integration: if  $f : \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}$  is log-concave, then

$$g(x) = \int f(x, y) \, dy$$

is log-concave (not easy to show)

# Log-Convex, Convex, Quasiconvex

log-convex  $\Rightarrow$  convex  $\Rightarrow$  quasiconvex

- $f$  is convex  $\Rightarrow f$  is quasiconvex;
- $f$  is log-convex  $\Rightarrow f$  is convex (because  $h$  is convex  $\Rightarrow e^h$  is convex).

concave  $\Rightarrow$  log-concave  $\Rightarrow$  quasiconcave

- $f$  is log-concave  $\Rightarrow f$  is quasiconcave: log function is monotone;
- $f$  is concave  $\Rightarrow f$  is log-concave (because  $h$  is concave  $\Rightarrow \log h$  is concave).

# Convexity and Continuity

The two concepts are closely related:

- Convex function  $f$  on  $\mathbf{R}^n$  is continuous relative to any relatively open convex set  $C$  in its effective domain.
- A convex function may be discontinuous at the relative boundary.

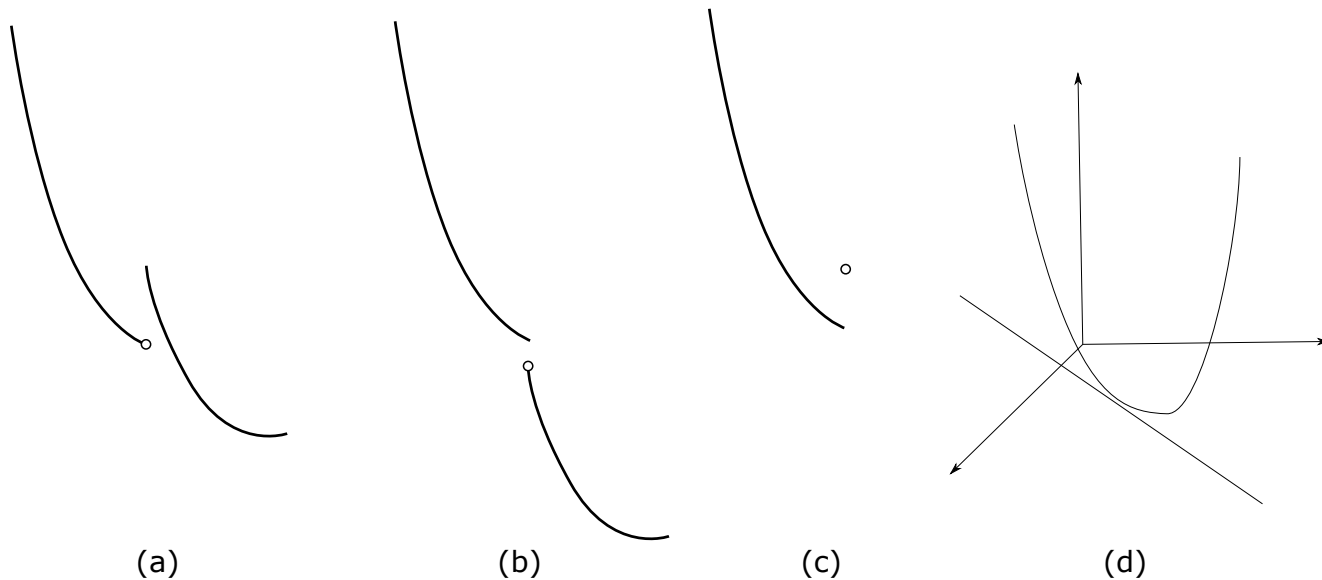


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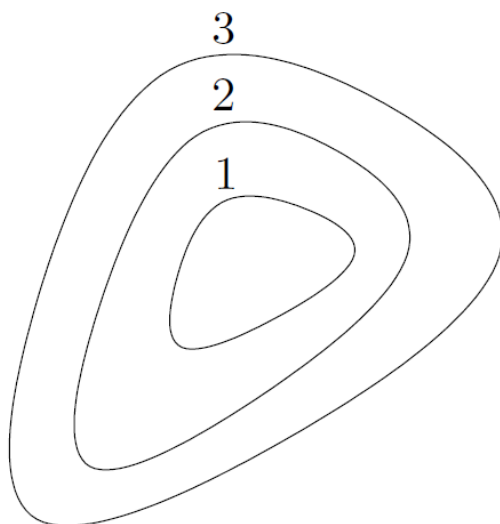
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“relative open set” and “relative boundary” require definitions of “affine hull”, “affine dimension”, and “relative interior”; we only illustrate with examples.



## Exercise 1

The level set of of a function is given as  $f(x) = \alpha$ . Is the following function quasi-convex, convex, concave, or quasi-concave?



## Exercise 2

Suppose  $f : \mathbf{R} \rightarrow \mathbf{R}$  is increasing and convex on its domain  $(a, b)$ . Let  $g$  denote its inverse. Is  $g(\cdot)$  a convex function, a concave function, or both, or neither?

## Exercise 3

Are the following functions convex, concave, quasiconvex, or quasiconcave?

1.  $e^x - 1$  on  $\mathbf{R}$ ;
2.  $f(x_1, x_2) = x_1 x_2$  on  $\mathbf{R}_{++}^2$ .

## Exercise 4

Show the following functions are convex

1.  $T(x, \omega) = x_1 + x_2 \cos \omega + x_3 \cos 2\omega + \dots + x_n \cos(n-1)\omega$ , and

$$f(x) = - \int_0^{2\pi} \log T(x, \omega) d\omega,$$

on  $\{x \in \mathbf{R}^n \mid T(x, \omega) > 0, 0 \leq \omega \leq 2\pi\}$ .

2.  $f(x) = -\log \det(A_0 + x_1 A_1 + x_2 A_2 + \dots + x_n A_n)$  where  $A_i \in \mathbf{S}^m$ , on  $\{x \mid A_0 + x_1 A_1 + x_2 A_2 + \dots + x_n A_n \succ 0\}$
3.  $f(x) = \sum_{i=1}^r |x|_{[i]}$  where  $|x|_{[i]}$  is the  $i$ -th largest absolute value in the components of the vector  $x$ .
4.  $f(x) = -\log(-\log(\sum_{i=1}^m e^{a_i^T x + b_i}))$  on  $\{x \mid \sum_{i=1}^m e^{a_i^T x + b_i} < 1\}$ .